

## SALTILLO, Mexico

WMO: 763900

Lat: 25.3514N

Lon: 101.0239W

Elev: 5873

StdP: 11.83

Time Zone: -6.00 (MEX)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |             |             |             |             |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|-------------|-------------|-------------|-------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |             |             | 99%         |             |             | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR          | MCDB        | DP          | HR          | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>29.9 | (c)<br>33.5 | (d)<br>15.3                   | (e)<br>14.8 | (f)<br>49.5 | (g)<br>19.6 | (h)<br>18.3 | (i)<br>47.9 | (j)<br>21.9           | (k)<br>68.1 | (l)<br>18.8 | (m)<br>62.4 | (n)<br>4.0            | (o)<br>210 | (p)<br>0.429 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD<br>to 0.4% DB |          |
|---------------|---------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|-------------------------|----------|
|               |                           | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                         |          |
|               |                           | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                    | PCWD     |
| (a)<br>6      | (b)<br>25.1               | (c)<br>93.3     | (d)<br>67.7 | (e)<br>91.0 | (f)<br>67.6 | (g)<br>88.2 | (h)<br>67.0 | (i)<br>72.8         | (j)<br>87.0 | (k)<br>71.4 | (l)<br>86.0 | (m)<br>69.9 | (n)<br>84.5 | (o)<br>10.5             | (p)<br>0 |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 68.3                            | 129.6 | 82.6 | 66.5 | 122.0 | 80.6 | 64.8 | 114.7 | 78.1 | 40.8          | 87.7 | 39.4 | 86.2 | 37.8 | 84.6 | 82.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |           |            |           |            |           |            |           |    |    |
|-------------------|----------|----------|----------------------------|----------|--------------------|---------|----------------------------------------------------|-----------|------------|-----------|------------|-----------|------------|-----------|----|----|
|                   |          |          | Mean                       |          | Standard Deviation |         | n=5 years                                          |           | n=10 years |           | n=20 years |           | n=50 years |           |    |    |
| 1%                | 2.5%     | 5%       | Min                        | Max      | Min                | Max     | Min                                                | Max       | Min        | Max       | Min        | Max       | Min        | Max       | DB | WB |
| (a) 20.8          | (b) 18.3 | (c) 16.6 | (e) 23.2                   | (f) 98.8 | (g) 4.8            | (h) 3.0 | (i) 19.8                                           | (j) 101.0 | (k) 17.0   | (l) 102.7 | (m) 14.3   | (n) 104.4 | (o) 10.9   | (p) 106.7 |    |    |
| (a) 20.8          | (b) 18.3 | (c) 16.6 | (e) 21.6                   | (f) 75.6 | (g) 4.0            | (h) 2.6 | (i) 18.7                                           | (j) 77.5  | (k) 16.4   | (l) 79.1  | (m) 14.1   | (n) 80.5  | (o) 11.2   | (p) 82.4  |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 65.3   | 54.3  | 58.3  | 62.1  | 68.3  | 72.7  | 73.3  | 73.1  | 72.4  | 68.6  | 65.3  | 59.7  | 55.5  |
|                                                                        | DBStd       | 9.57   | 8.83  | 8.79  | 8.20  | 7.48  | 6.24  | 3.88  | 3.31  | 3.44  | 4.24  | 6.20  | 7.21  | 8.87  |
|                                                                        | HDD50       | 169    | 62    | 28    | 12    | 2     | 0     | 0     | 0     | 0     | 0     | 2     | 11    | 52    |
|                                                                        | HDD65       | 1343   | 338   | 212   | 148   | 53    | 13    | 0     | 0     | 1     | 17    | 71    | 185   | 305   |
|                                                                        | CDD50       | 5767   | 196   | 261   | 387   | 551   | 705   | 698   | 716   | 695   | 557   | 476   | 302   | 223   |
|                                                                        | CDD65       | 1463   | 7     | 24    | 58    | 152   | 253   | 248   | 251   | 231   | 124   | 79    | 26    | 10    |
|                                                                        | CDH74       | 14684  | 161   | 393   | 778   | 1717  | 2628  | 2452  | 2159  | 2046  | 1031  | 767   | 347   | 205   |
|                                                                        | CDH80       | 5171   | 23    | 94    | 214   | 704   | 1159  | 945   | 744   | 705   | 273   | 212   | 70    | 28    |
| (14) Wind                                                              | WSAvg       | 7.2    | 6.3   | 7.0   | 7.9   | 8.5   | 8.6   | 8.3   | 7.8   | 7.3   | 6.4   | 6.3   | 6.3   | 6.1   |
| (15) Precipitation                                                     | PrecAvg     | 18.4   | 0.6   | 0.5   | 0.6   | 0.5   | 1.6   | 2.3   | 3.9   | 3.1   | 3.1   | 1.2   | 0.5   | 0.5   |
|                                                                        | PrecMax     | 31.2   | 4.3   | 1.8   | 4.6   | 1.5   | 2.9   | 4.2   | 12.0  | 7.2   | 8.2   | 4.3   | 3.2   | 2.6   |
|                                                                        | PrecMin     | 8.7    | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 1.0   | 0.8   | 0.2   | 0.0   | 0.0   | 0.0   |
|                                                                        | PrecStd     | 6.2    | 0.8   | 0.6   | 0.9   | 0.4   | 0.9   | 1.2   | 2.6   | 1.6   | 2.0   | 1.2   | 0.7   | 0.7   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 82.6  | 87.6  | 89.2  | 95.1  | 98.3  | 95.3  | 92.9  | 91.7  | 89.7  | 89.5  | 86.1  | 82.7  |
|                                                                        |             | MCWB   | 60.2  | 61.9  | 63.0  | 66.5  | 68.7  | 69.2  | 70.9  | 70.2  | 69.9  | 68.3  | 64.5  | 60.9  |
|                                                                        | 2%          | DB     | 78.5  | 82.6  | 85.8  | 91.3  | 93.4  | 91.4  | 89.6  | 89.4  | 86.1  | 85.8  | 82.1  | 79.1  |
|                                                                        |             | MCWB   | 58.3  | 60.5  | 62.1  | 64.9  | 66.6  | 68.8  | 69.6  | 69.8  | 67.7  | 67.1  | 62.6  | 59.5  |
|                                                                        | 5%          | DB     | 74.9  | 78.9  | 82.3  | 87.7  | 89.8  | 88.2  | 86.4  | 86.3  | 82.7  | 82.0  | 78.0  | 75.5  |
|                                                                        |             | MCWB   | 56.7  | 58.5  | 60.2  | 63.8  | 66.1  | 67.5  | 68.1  | 68.6  | 66.7  | 64.8  | 60.3  | 57.4  |
|                                                                        | 10%         | DB     | 70.2  | 75.0  | 78.6  | 84.1  | 86.4  | 85.9  | 84.4  | 84.2  | 80.4  | 78.4  | 73.9  | 71.5  |
|                                                                        |             | MCWB   | 54.3  | 56.7  | 58.5  | 62.2  | 65.1  | 67.4  | 67.0  | 67.4  | 66.0  | 62.7  | 58.6  | 55.8  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 63.2  | 65.6  | 67.4  | 72.9  | 73.3  | 75.0  | 73.7  | 73.7  | 72.9  | 71.5  | 68.1  | 64.5  |
|                                                                        |             | MCDB   | 78.2  | 81.0  | 83.3  | 88.2  | 89.6  | 87.3  | 87.5  | 87.2  | 86.2  | 86.6  | 81.2  | 78.0  |
|                                                                        | 2%          | WB     | 60.1  | 62.3  | 64.4  | 68.3  | 70.8  | 72.0  | 71.8  | 71.7  | 70.4  | 68.3  | 64.4  | 61.3  |
|                                                                        |             | MCDB   | 75.9  | 79.0  | 81.4  | 86.2  | 87.6  | 86.9  | 85.9  | 85.9  | 82.5  | 81.6  | 77.6  | 76.6  |
|                                                                        | 5%          | WB     | 57.9  | 59.7  | 62.1  | 66.1  | 68.6  | 69.9  | 70.1  | 70.1  | 68.4  | 66.2  | 62.0  | 58.9  |
|                                                                        |             | MCDB   | 72.9  | 76.7  | 79.0  | 83.0  | 85.8  | 85.4  | 84.2  | 84.3  | 79.8  | 78.2  | 74.9  | 73.6  |
|                                                                        | 10%         | WB     | 55.0  | 57.7  | 59.7  | 64.0  | 66.6  | 68.2  | 68.3  | 68.3  | 66.9  | 64.4  | 59.7  | 56.4  |
|                                                                        |             | MCDB   | 68.8  | 72.9  | 76.1  | 81.1  | 82.9  | 83.0  | 81.9  | 81.9  | 77.4  | 75.8  | 72.0  | 69.7  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 26.4  | 26.9  | 27.3  | 27.1  | 26.6  | 25.1  | 24.0  | 24.6  | 23.3  | 25.2  | 25.8  | 25.9  |
|                                                                        |             | MCDBR  | 30.6  | 31.3  | 31.2  | 31.3  | 29.4  | 28.7  | 27.0  | 28.1  | 27.8  | 30.3  | 30.4  | 30.1  |
|                                                                        |             | MCWBR  | 19.5  | 19.3  | 18.7  | 18.7  | 16.7  | 16.4  | 15.2  | 15.4  | 15.1  | 17.4  | 19.0  | 18.9  |
|                                                                        | 5% WB       | MCDBR  | 29.4  | 30.1  | 29.5  | 28.6  | 27.2  | 26.4  | 25.8  | 26.6  | 25.5  | 26.8  | 27.7  | 28.9  |
|                                                                        |             | MCWBR  | 19.8  | 19.7  | 19.1  | 19.3  | 17.2  | 16.7  | 15.8  | 15.7  | 15.0  | 16.4  | 18.2  | 19.1  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.271 | 0.281 | 0.298 | 0.331 | 0.349 | 0.342 | 0.347 | 0.330 | 0.337 | 0.308 | 0.289 | 0.274 |
|                                                                        | taud        |        | 2.594 | 2.549 | 2.500 | 2.403 | 2.392 | 2.480 | 2.498 | 2.559 | 2.516 | 2.581 | 2.571 | 2.596 |
|                                                                        | Ebn at Noon |        | 318   | 320   | 318   | 306   | 298   | 298   | 297   | 303   | 300   | 306   | 308   | 312   |
|                                                                        | Edh at Noon |        | 27    | 30    | 34    | 38    | 39    | 35    | 35    | 32    | 33    | 29    | 28    | 26    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1306  | 1606  | 1904  | 2122  | 2161  | 2159  | 2052  | 2018  | 1708  | 1629  | 1390  | 1237  |
|                                                                        | RadStd      |        | 108   | 162   | 134   | 126   | 88    | 142   | 140   | 107   | 151   | 125   | 128   | 91    |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (3 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | -1.78 | -1.01       | N/A   | N/A   | N/A   |

Nomenclature: See separate page